How can 3D-based interactive visual design enhance descriptive thinking and visual awareness in adults?

University of Winchester

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Contents

1. Abstract

Many adults engage regularly with visual content yet struggle to describe what they see beyond surface-level observations. While existing digital cognitive training tools often focus on memory, speed, or attention, there is limited research into how interactive design can support descriptive thinking and visual awareness. This study investigates the question: How can 3D-based interactive visual design enhance descriptive thinking and visual awareness in adults?

A mixed-method design-led approach was adopted, combining a literature review with an online survey involving two visual description tasks and feature preference questions. The findings reveal that participants consistently prioritised immediately salient visual elements such as colour and lighting, while spatial relationships, function, and narrative meaning were less frequently articulated. Structured visual environments were easier to describe than complex scenes, highlighting the role of visual hierarchy and guided attention.

Based on these insights, a conceptual interactive artefact was developed to demonstrate how guided prompts, visual emphasis, and vocabulary support could enhance descriptive engagement. The study contributes to design research by proposing an alternative approach to visual training tools that prioritises reflective observation and articulation, providing a foundation for prototype development in a subsequent practical phase.

2. Introduction

2.1 Personal Motivation

This research topic emerged from my own experience of finding it difficult to describe what I see in a clear and detailed way. Although I often think deeply about visual details, I realised that my verbal descriptions tended to remain broad or general. This limitation became more visible during my university studies, where reflective communication and visual explanation are essential skills. Recognising this gap led me to explore whether descriptive thinking can be strengthened through intentional practice, and whether interactive 3D environments might support adults in developing greater visual awareness. This personal insight shaped the direction of the project and informed the development of the research question.

2.1 Background and Context

In everyday life, adults constantly interact with visual environments such as interiors, urban spaces, digital media, and imagery. While many people can recognise whether a visual scene feels appealing or engaging, they often struggle to articulate why. Descriptions frequently remain limited to broad terms such as “nice” or “beautiful,” without deeper reference to visual elements such as colour, lighting, texture, spatial layout, or atmosphere. This difficulty can limit communication, creative expression, and visual literacy.

Visual awareness and descriptive thinking play an important role in how individuals interpret and communicate their experiences of the world. Research in cognitive psychology and design theory suggests that perception and verbalisation are closely related but do not automatically develop together (Peelen, 2024). Being able to see does not necessarily mean being able to describe, interpret, or reflect on visual information in depth (Lamme, 2020). As a result, there is growing interest in tools and methods that actively support observational and descriptive skills rather than assuming these abilities improve naturally through exposure alone.

2.2 Research Problem and Gap

In recent years, many digital tools have been developed to support cognitive training. Popular applications and brain-training platforms often focus on improving memory, reaction speed, attention, or problem-solving performance. While these tools may enhance certain cognitive abilities, they rarely address descriptive thinking or visual articulation as core skills. Most prioritise efficiency and performance metrics rather than reflective observation or expressive engagement with visual content.

At the same time, research in design cognition and visual perception highlights the importance of describing visual information as a way of deepening understanding and supporting creativity. However, there is limited research exploring how interactive digital design—particularly 3D-based visual environments—can be used to support descriptive thinking and visual awareness in adult users outside formal education or clinical contexts. This gap indicates a need for exploratory research into interactive visual systems that encourage users to pause, observe, and describe what they see rather than react quickly or complete tasks under time pressure.

2.3 Research Question and Aims

In response to this gap, this study investigates the following research question:

How can 3D-based interactive visual design enhance descriptive thinking and visual awareness in adults?

The study is guided by three aims:

1. To analyse existing theories and studies on descriptive thinking, visual awareness, and 3D interaction design.
2. To investigate how adults perceive and describe visual details within 3D-based visual contexts.
3. To translate findings into design ideas that guide prototype development for an interactive visual learning app in Term 2.

2.4 Research Approach and Structure

To address the research question, the study adopts a qualitative, exploratory approach using an online survey that combines background questions, visual description tasks, and design feature preference questions. This method enables participants to express how they perceive and describe visual scenes in their own words, providing insight into descriptive patterns and challenges.

The dissertation is structured as follows: the literature review examines theoretical perspectives on visual perception and descriptive thinking; the methodology outlines the research design and data collection process; the artefact section presents the conceptual design outcomes; the evaluation and discussion interpret the findings; and the conclusion reflects on the study’s contributions, limitations, and future development.

3. Literature Review

3.1 Introduction to the Literature Review

The purpose of this literature review is to examine existing research related to visual perception, descriptive thinking, visual awareness, and interactive digital design, with a particular focus on how these areas intersect within 3D-based visual environments. The review establishes a theoretical foundation for the study and identifies a gap in current research concerning tools that actively support descriptive thinking in adults. By analysing literature from cognitive psychology, design theory, and human–computer interaction, this section situates the research question within existing academic discourse.

3.2 Visual Perception and Visual Thinking

Research in cognitive psychology suggests that perception is not a passive process, but an active construction of meaning. Arnheim (1969) argues that visual perception and thinking are inseparable, proposing that humans "think visually" by organising and interpreting what they see. This theory highlights the importance of developing visual awareness as a cognitive skill rather than assuming it develops automatically.

Tversky (2011) further supports this view, emphasising that perception, language, and spatial reasoning are deeply interconnected. She suggests that the ability to describe visual information relies on both perceptual sensitivity and conceptual understanding. These findings are directly relevant to this study, as they suggest that difficulty in describing visuals may stem from cognitive processes rather than a lack of exposure to visual stimuli.

Goldstein (2014) expands on this by explaining how attention, salience, colour, and contrast influence what individuals notice first within a scene. This research helps explain why participants in the present study consistently focused on colour and lighting when describing visual scenes.

3.3 Descriptive Thinking and Design Cognition

Within design research, descriptive thinking is closely linked to creativity and reflective practice. Lawson (2004) explains that designers develop understanding through observing, describing, and reinterpreting visual information. The act of verbalising visual elements allows individuals to move beyond surface-level perception and engage in deeper cognitive processing.

Goldschmidt (2017) introduces the concept of "design cognition," arguing that description plays a critical role in how designers explore and develop ideas. According to Goldschmidt, the ability to articulate visual observations supports both creative reasoning and problem-solving. Although her research focuses on designers, the principles are applicable to adults more broadly, suggesting that descriptive skills can be trained and enhanced.

These studies support the argument that descriptive thinking is a learnable skill and provide theoretical justification for exploring interactive tools that encourage structured observation and articulation.

3.4 Visual Awareness, Attention, and Cognitive Load

Research into attention and cognitive load provides further insight into why some visual scenes are easier to describe than others. Sweller (1988) introduces Cognitive Load Theory, explaining that complex environments can overwhelm working memory, reducing clarity of thought and expression.

Norman (2013) applies these principles to design, arguing that well-structured visual systems reduce cognitive load and support understanding. His work suggests that design features such as hierarchy, feedback, and guidance can improve user engagement and comprehension.

This aligns with the findings of the present study, where participants found structured interior scenes easier to describe than complex urban environments. These theories help explain how interactive visual design could guide attention and support descriptive clarity.

3.5 3D Visualisation and Spatial Cognition

3D visualisation has been shown to support spatial understanding and engagement. Billinghurst and Duenser (2012) demonstrate that interactive 3D environments can enhance spatial cognition by allowing users to explore depth, scale, and relationships between objects.

Although much research in this area focuses on education or augmented reality, the underlying principles remain relevant. The ability to navigate and interpret 3D space may strengthen visual awareness and imaginative engagement. This supports the study's focus on 3D-based visual design as a foundation for developing descriptive thinking.

However, existing research often prioritises task performance or learning outcomes rather than reflective description, highlighting a gap that this study seeks to address.

3.6 Digital Interaction, UX, and Reflective Engagement

Human–computer interaction research emphasises the importance of designing systems that support reflection rather than speed alone. Shneiderman (2022) argues for human-centred digital systems that promote understanding, creativity, and agency. He suggests that interactive tools should encourage users to pause, reflect, and make meaning from content.

Similarly, Norman (2013) highlights the role of feedback and affordances in helping users understand what they are seeing and doing. These ideas directly inform the design feature preferences explored in this study, such as guided prompts, visual highlighting, and vocabulary support.

3.7 Limitations of Existing Cognitive Training Tools

Many commercial brain-training apps focus on improving memory, attention, and processing speed rather than reflective or descriptive skills (Bilodeau, 2022). Popular platforms such as Lumosity and CogniFit emphasise rapid cognitive exercises and performance metrics, offering little support for visual articulation or descriptive awareness. Recent research by Abd-alrazaq et al. (2023) similarly shows that serious games can enhance attention in cognitively impaired adults, but again prioritise performance outcomes over expressive or reflective engagement. Together, these findings reinforce the gap identified in this study: the lack of tools designed to support descriptive visual awareness in adults.

3.8 Research Gap and Relevance to the Study

The reviewed literature demonstrates strong theoretical support for the importance of visual perception, descriptive thinking, and structured interaction. However, there is limited research exploring how 3D-based interactive visual design can be used specifically to train descriptive thinking and visual awareness in adults outside of formal education or clinical contexts.

This study responds to that gap by combining theories of visual cognition, design thinking, and digital interaction with empirical user data. The literature reviewed here provides a foundation for analysing the survey findings and informing the development of a prototype in the subsequent practical phase of the project.

3.9 Linking Literature Review to Methodology

The literature indicates that descriptive thinking and visual awareness are subjective, interpretive processes that require methods capable of capturing personal perception and expression (Arnheim, 1969; Lawson, 2004). For this reason, a qualitative, survey-based methodology was selected, combining open-ended visual description tasks with structured questions. This approach allows participants to articulate how they interpret visual scenes, while drawing on principles from visual cognition, spatial visualisation, and user-centred design to inform prototype development.

4. Methodology

4.1 Research Approach

This study adopts a qualitative-dominant mixed-method approach, combining qualitative and descriptive quantitative data. This approach is appropriate because the research investigates how adults perceive and describe visual scenes, rather than measuring performance or testing hypotheses. The emphasis is on understanding participant experiences, language use, and descriptive patterns in response to visual stimuli.

Quantitative elements, such as Likert-scale responses, are used to provide supporting context and identify general trends (for example, perceived ease of description or interest in design features). However, the core of the analysis focuses on qualitative written responses, which allow deeper insight into descriptive thinking and visual awareness. This approach aligns with an interpretivist perspective, where meaning is constructed through participants' subjective experiences.

4.2 Research Method

The primary research method was an online survey, designed to be accessible, time-efficient, and suitable for exploratory research. The survey was distributed digitally using Google Forms and consisted of four sections: background questions, two visual description tasks, and design feature preference questions.

The survey included:

- Closed-ended questions to gather contextual and self-reported data (e.g. confidence levels and frequency of visual engagement).
- Open-ended descriptive tasks in which participants were asked to describe visual scenes in their own words.
- Likert-scale questions to explore perceived ease of description and the usefulness of proposed interactive features.

This method was selected because it allowed participants to engage with visual material independently and respond in their own time, reducing pressure and supporting reflective observation. It also enabled the collection of both measurable trends and rich descriptive data within a single instrument.

4.3 Participants and Sampling

A convenience sampling strategy was used to recruit adult participants for the study. A total of eight participants completed the survey. Although the sample size was small, it was appropriate for the exploratory nature of the research and enabled detailed qualitative analysis of individual responses. Participants were not required to have a background in design or visual disciplines, aligning with the study's aim to investigate visual awareness and descriptive thinking in a general adult population.

Participants were recruited primarily through email outreach and internal university channels. Invitations were sent via university email to external contacts connected to previous projects, asking recipients to share the survey within their organisation. The survey link was also shared in the Microsoft Teams channel used for the research module, enabling classmates to participate. Personal networks were used later to support response rates once initial participation was lower than anticipated. This combination of recruitment routes supported a practical and transparent approach to sampling for a small-scale, exploratory study.

4.4 Visual Stimuli

Two static 3D visual scenes were used as stimuli within the survey:

- Visual Task 1 presented an interior scene featuring an open-plan living, dining, and kitchen space with strong colour contrast, natural lighting, and clear spatial organisation.
- Visual Task 2 presented an urban environment incorporating roads, architectural elements, lighting conditions, and a monorail system, resulting in a more complex visual composition.

The use of two contrasting scenes supported comparison between structured and visually complex environments. This enabled analysis of how scene composition influences descriptive depth, attention patterns, and perceived ease of description. The visuals were presented as static images to ensure that all participants viewed the same content under consistent conditions.

4.5 Data Collection Procedure

Participants first read a short introduction explaining the purpose of the study and confirming anonymity and voluntary participation. After providing informed consent, they completed the survey in the following sequence:

1. Background questions relating to visual engagement and confidence.
2. Visual Task 1, including descriptive and reflective questions.
3. Visual Task 2, using a similar structure.
4. Questions exploring interest in potential design features and app usage.

The survey took approximately 8–10 minutes to complete. No personally identifiable data was collected.

4.6 Data Analysis

Data analysis was conducted in two stages:

- Quantitative data (Likert-scale and multiple-choice responses) were summarised using frequencies and percentages to identify trends in confidence, ease of description, and feature preferences.
- Qualitative data from open-ended responses were analysed thematically. Responses were reviewed and coded to identify recurring descriptive patterns, such as focus on colour, lighting, materials, spatial layout, emotional interpretation, or meaning.

The qualitative analysis focused on how participants structured their descriptions, what visual elements they prioritised, and how these differed between the two scenes. These insights were then interpreted in relation to the research question and relevant literature.

4.7 Ethical Considerations

Ethical approval was obtained prior to data collection. Participation was voluntary, and participants were informed that they could withdraw at any time before submitting their responses. All data was collected anonymously and stored securely in accordance with university guidelines and data protection regulations.

As the study involved only non-sensitive visual material and reflective description tasks, no significant ethical risks were identified. Participants were not exposed to distressing content, and no deception was involved.

4.8 Methodological Limitations

The study has several limitations. The small sample size limits the generalisability of the findings, and the use of convenience sampling may introduce selection bias, as participants may already have an interest in visual content. Additionally, the use of static images does not fully replicate interactive visual experiences, which may affect how participants engage with the scenes.

However, these limitations are acceptable within the context of an exploratory design-led study. The methodology successfully generated meaningful insights that inform both the research question and the planned prototype development in Term 2.

5. Artefact / Practice

5.1 Purpose of the Artefact

The practical outcome of this research is the conceptual development of an interactive visual learning tool informed by the study's findings. Rather than producing a fully developed application, the artefact functions as a research-informed prototype concept that demonstrates how interactive visual design could support descriptive thinking and visual awareness in adults.

The artefact translates theoretical insights and participant responses into practical design principles that can be tested and refined in Term 2. It serves as a link between research and practice by showing how concepts from visual perception, design cognition, and user experience can be applied within an interactive framework.

5.2 Design Rationale Informed by Research Findings

Survey findings revealed that although participants frequently engage with visual content, their confidence and descriptive depth varied, particularly when scenes became more complex. Participants tended to focus on immediately salient elements such as colour and lighting, while spatial relationships, function, and narrative meaning were less frequently addressed.

The artefact is therefore designed around guided observation. Its purpose is to encourage users to move beyond first impressions and engage more reflectively with visual scenes. Design principles derived from the research include:

- **Prompted Attention:** Guided questions encourage users to notice overlooked visual elements.
- **Visual Emphasis:** Highlighting or zooming directs attention without overwhelming the user.
- **Vocabulary Support:** Word suggestions assist users in articulating what they perceive.
- **Low Cognitive Pressure:** The absence of time limits prioritises reflection over performance.

These principles respond directly to participant preferences and address limitations observed in existing cognitive training tools.

5.3 Intended Format and Interaction Model

The artefact is intended as a mobile-based interactive prototype designed for short, everyday use. Users would be presented with selected 3D visual scenes, such as interiors or urban environments, which function as visual stimuli rather than navigable spaces.

Interaction follows a simple three-stage structure:

1. **Initial Observation:** Users describe what they notice first.
2. **Guided Exploration:** Prompts encourage attention to specific visual aspects.
3. **Reflection:** Users review and reconsider their descriptions.

This structure reflects survey findings, where structured scenes supported clearer description than visually complex environments, and allows complexity to be adjusted for different users.

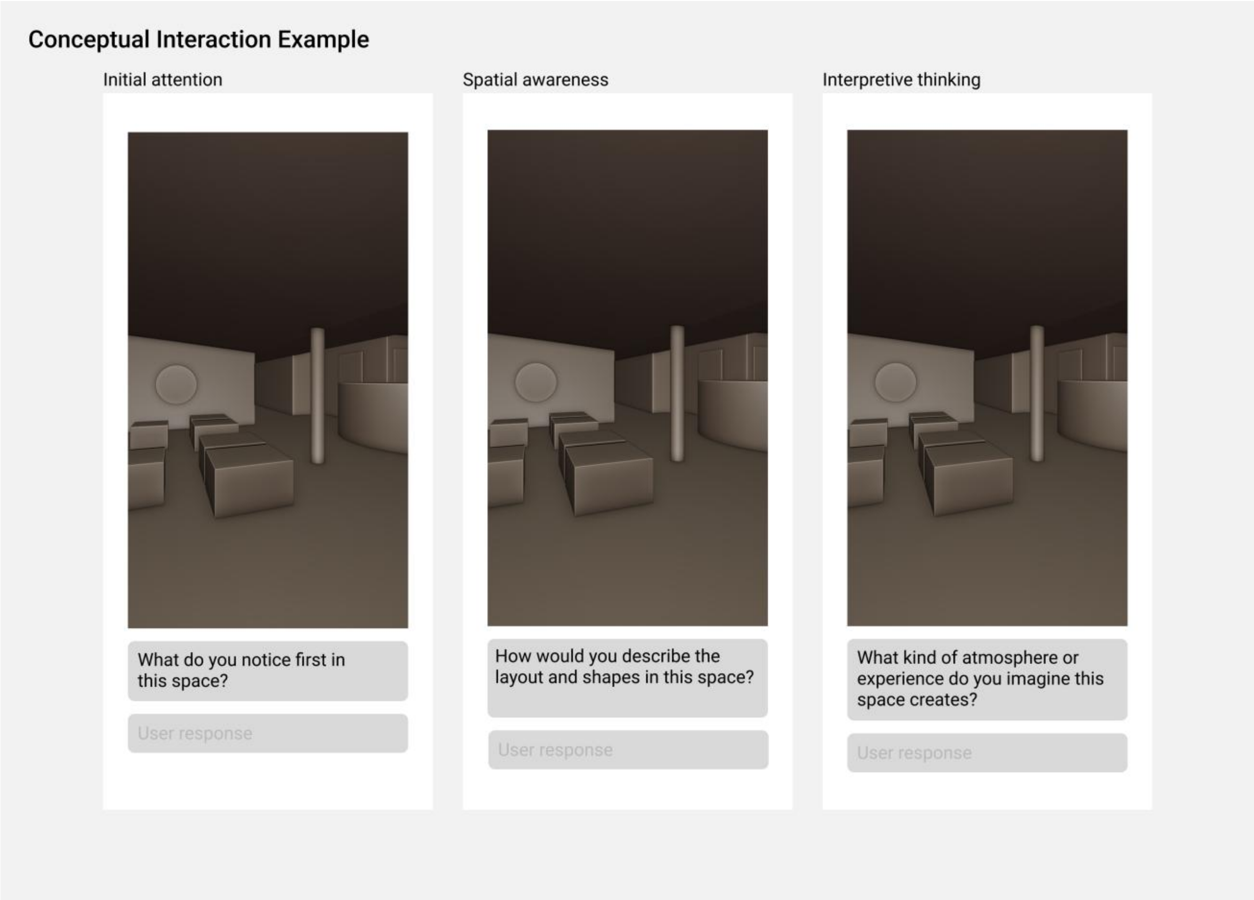


Figure 1: Conceptual Interaction Example Showing 3 Stages of Guided Visual Engagement

To illustrate how the research findings translate into an interactive format, an initial concept sketch was produced using a simple 3D scene from a real client project. The same visual stimulus is paired with three different guided prompts—Initial Attention, Spatial Awareness, and Interpretive Thinking—to demonstrate how users might engage with descriptive thinking at different levels. This sketch is not a design prototype but a research artefact used to explore interaction principles for development in Term 2.

5.4 Relationship to Term 2 Development

In Term 2, the artefact will be developed as a low- to medium-fidelity prototype supported by sketches, interface layouts, and interaction flows. The focus will remain on demonstrating how research findings inform design decisions rather than technical implementation.

5.5 Contribution to Practice

The artefact proposes an alternative approach to visual training tools by prioritising descriptive awareness and reflective engagement rather than performance metrics. It demonstrates how interactive 3D visual design can support cognitive and expressive skills, offering value to fields such as visual design, education, and user experience.

6. Evaluation & Discussion

6.1 Overview of Findings

This section evaluates the results of the online survey, which explored how adults notice and describe visual scenes and how interactive design features might support descriptive thinking and visual awareness. Although the sample size was small (n = 8), the responses offer useful insight into how adults engage with visual content and where design interventions may be beneficial. The discussion connects these findings to the research question and relevant literature.

6.2 Participant Background and Self-Perceived Ability

Five participants reported engaging with visually focused activities very often, suggesting regular exposure to visual material. However, confidence levels varied: three participants felt very confident describing visuals, while others reported only slight or moderate confidence.

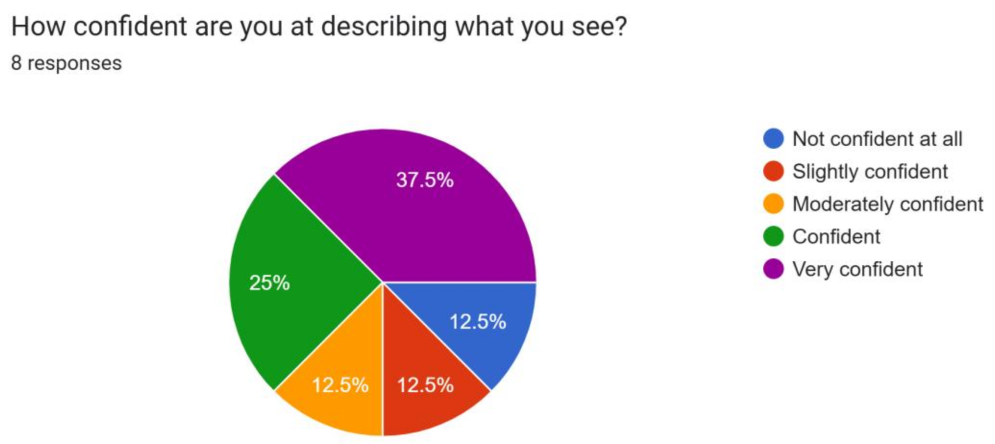


Figure 2: Participant confidence levels in describing visual scenes.

This contrast suggests that exposure alone does not guarantee strong descriptive ability. This supports the literature arguing that perception and articulation are related but distinct processes (Arnheim, 1969; Tversky, 2011). The finding also reinforces the need for tools that support the transition from noticing to verbalising.

6.3 Visual Task 1: Interior Scene

6.3.1 Initial Attention and Visual Salience

Participants showed strong consistency in what they noticed first. Four participants immediately focused on the green sofas, with others mentioning sunlight or the dining table. Colour and lighting clearly acted as focal points.

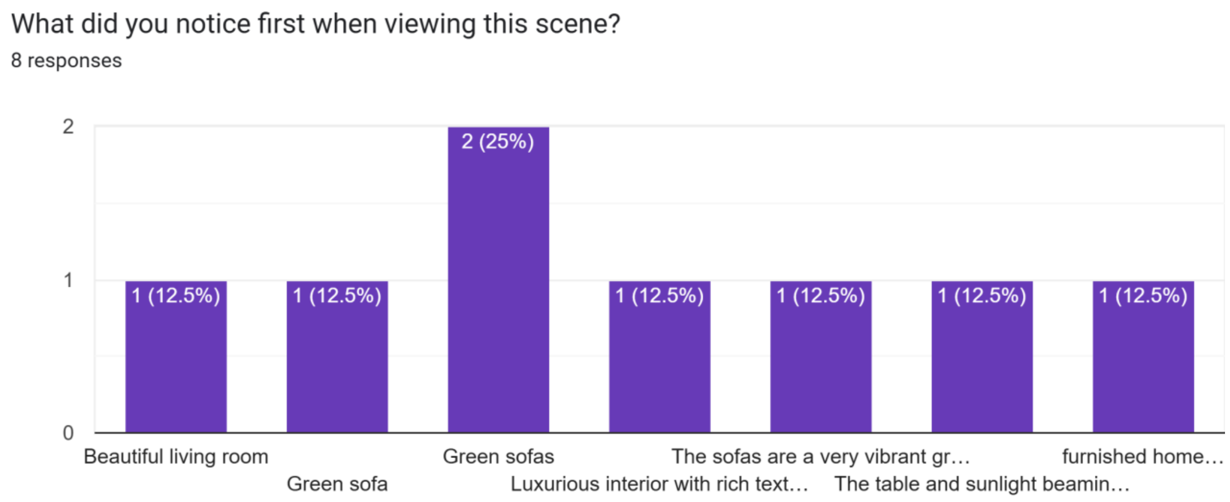


Figure 3: Initial attention patterns in the interior scene.

This aligns with Goldstein's (2014) work on stimulus salience, where contrast and illumination guide early attention. The interior scene's clear hierarchy supported both noticing and describing.

6.3.2 Descriptive Depth and Visual Awareness

Descriptions of the interior scene were generally detailed. Participants referenced materials (marble, wood), layout (open-plan), and atmosphere (warm, modern, nature-inspired). Several comments moved beyond listing objects and interpreted mood or theme.

This supports Lawson (2004) and Goldschmidt (2017), who argue that description deepens when viewers recognise relationships between form, material, and function. The structured scene provided strong cues for reflective observation.

6.3.3 Ease of Description and Focus Areas

Six participants found the interior scene easy to describe. Colour and lighting were the most frequently selected focus areas (both chosen by seven participants), while spatial depth and narrative meaning were mentioned less often.

This suggests adults naturally prioritise salient aesthetics over interpretive or relational details. For interactive tools, this indicates a need for prompts that broaden attention beyond colour and lighting.

6.4 Visual Task 2: Urban Scene

6.4.1 Initial Attention and Complexity

Responses to the second scene were more varied. Participants noticed different entry points — roads, monorail, skyline, lighting — with no single dominant focal point. This reflects the higher complexity of the scene.

Research on spatial cognition (Billinghurst & Duenser, 2012) supports the idea that complex environments increase cognitive load and make descriptive organisation more difficult.

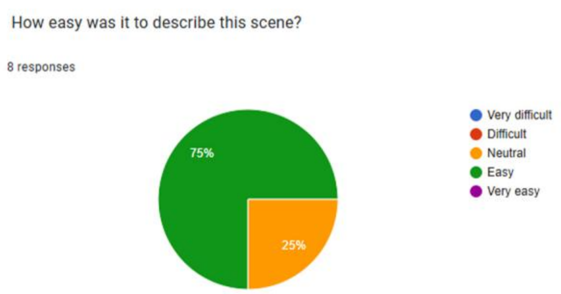
6.4.2 Descriptive Content and Interpretation

Descriptions of the urban scene were broader but less structured. Participants often listed elements rather than explaining relationships or layout. Several interpreted the scene as futuristic or industrial, showing imaginative engagement but less clarity.

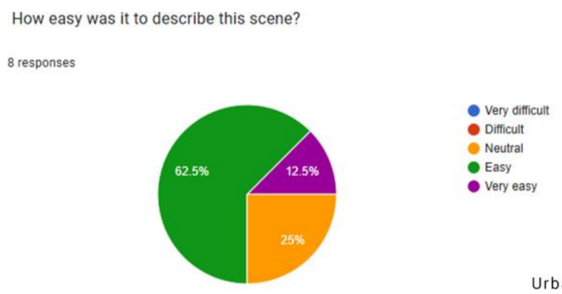
This suggests that while complex scenes may inspire interpretation, they can reduce descriptive organisation without guidance.

6.4.3 Ease of Description

Five participants found the scene easy to describe, two were neutral, and one found it very easy. Overall ease ratings were slightly lower than for the interior scene.



Interior Scene



Urban Scene

Figure 4: Comparison of ease ratings for both visual scenes.

This supports the idea that visual hierarchy and clarity help users describe scenes more confidently.

6.5 Design Feature Preferences and App Interest

Participants responded positively to features designed to support descriptive thinking. Guided prompts, visual highlighting, and vocabulary support received the strongest approval. These features directly address the main challenge observed: adults notice key elements but may struggle to articulate them fully.

Side-by-side comparisons and timed modes received more neutral responses, suggesting they may be useful only in specific contexts. This aligns with Norman's (2013) emphasis on flexibility and user control.

Interest in a mobile application was generally positive: seven out of eight participants selected "Yes" or "Maybe."

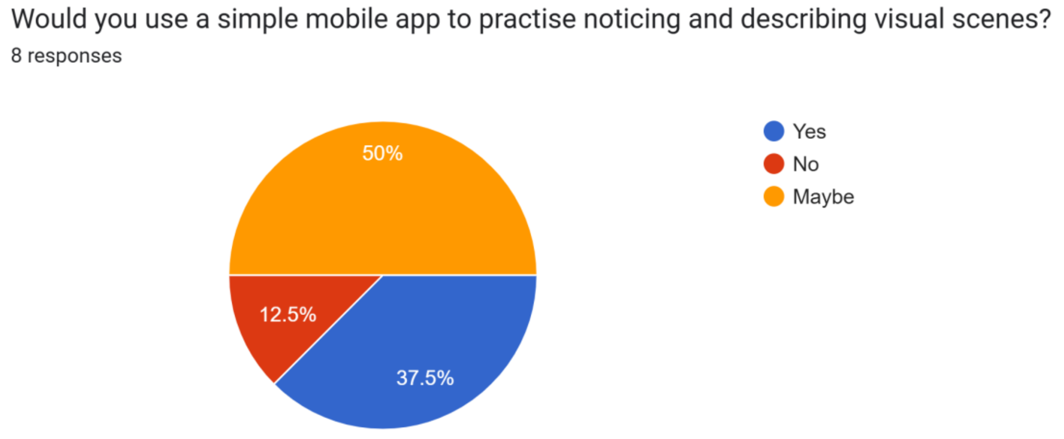


Figure 5: Responses to whether participants would use a descriptive-thinking app.

This indicates sufficient openness to justify developing a prototype in Term 2.

6.6 Discussion in Relation to the Research Question

The findings provide clear support for the research question: How can 3D-based interactive visual design enhance descriptive thinking and visual awareness in adults?

Key insights include:

- Adults notice visual information consistently, but descriptive confidence varies.
- Structured scenes with clear hierarchy support more detailed and organised descriptions.
- Complex scenes increase cognitive load and reduce descriptive clarity.
- Participants value features that guide attention, support vocabulary, and encourage reflection.
- Users prefer reflective, non-timed interaction over speed-based tasks.

Together, these findings suggest that a 3D-based interactive application incorporating prompts, highlighting, and vocabulary support could meaningfully enhance descriptive thinking by helping users notice, interpret, and articulate visual information more deeply.

7. Project Timeline

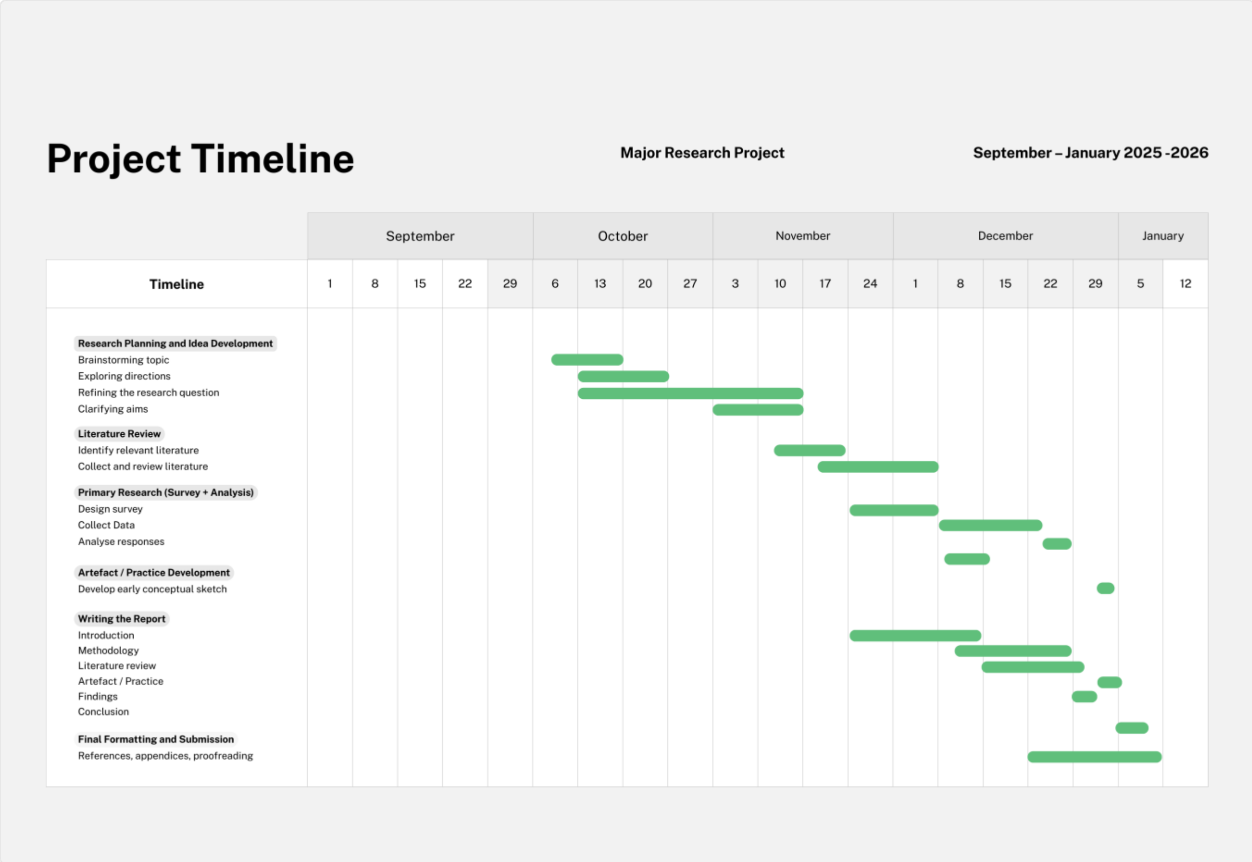


Figure 6: Project Timeline

The Semester 1 project timeline (Figure 6) summarises the structured progression of research, writing, and early artefact development undertaken throughout the term.

The project began with brainstorming and the exploration of possible directions, followed by the refinement of the research question and aims. An initial review of relevant literature was then undertaken to establish the theoretical foundation for the study. The survey design stage required an iterative process, with additional time allocated to refining the questions to ensure clarity, relevance, and appropriate length; during this period, the Introduction chapter was also drafted. Once the survey was launched, data collection took place and the Methodology chapter was developed in parallel. The Literature Review chapter was completed slightly later, drawing on the earlier reading. Once response levels stabilised, the survey was closed and the analysis was carried out. After this, an early conceptual artefact sketch was produced, which then informed the development of the accompanying theoretical and practice-based text outlining the planned approach for Semester 2. The final stages focused on drafting the remaining sections, completing the conclusion, and undertaking final formatting, proofreading, and the preparation of references and appendices ahead of submission.

8. Conclusion

This research explored the question: How can 3D-based interactive visual design enhance descriptive thinking and visual awareness in adults? Through a qualitative, design-led approach combining a literature review, an online survey, and conceptual artefact development, the study investigated how adults perceive and describe visual scenes and how interactive design features might support this process.

The literature review established that visual perception and descriptive thinking are active cognitive processes rather than automatic outcomes of visual exposure. Theories of visual thinking and design cognition highlight the importance of articulation and reflection in developing visual awareness, while research in user experience and cognitive load suggests that structured guidance can support understanding and engagement. Together, these perspectives revealed a gap in existing digital tools, which tend to prioritise speed or performance over descriptive awareness.

Survey findings supported these theoretical insights. Although participants often engaged with visual content, confidence in describing what they saw varied. Participants consistently focused on immediately salient elements such as colour and lighting, while spatial relationships, function, and narrative meaning were less frequently articulated. Structured interior scenes were generally easier to describe than complex urban environments, indicating the importance of visual hierarchy and clarity.

Participants also responded positively to proposed interactive features such as guided prompts, visual highlighting, and vocabulary support. These findings suggest that adults are receptive to tools that actively guide observation and articulation rather than testing cognitive performance alone. This demonstrates that 3D-based interactive visual design can enhance descriptive thinking when it supports attention, reflection, and expression.

The artefact developed in this study takes the form of an early conceptual sketch that visualises how the research findings could translate into an interactive format. It is not a design or functional prototype, but a simple representation used to explore how guided prompts might structure user engagement with a 3D visual scene. This sketch provides a clear starting point for future development and demonstrates how the insights gained in Term 1 can inform the direction of the prototype work planned for Term 2.

While the study is limited by a small participant sample and the use of static visual stimuli, these constraints are appropriate for an exploratory design-led investigation. In Term 2, the research will be extended through the development and evaluation of a prototype informed by the findings presented here.

Overall, this study contributes to design-led research by demonstrating how interactive 3D visual design can support descriptive thinking and visual awareness in adults, offering an alternative approach to digital visual training that prioritises reflective engagement over performance metrics.

9. AI Use Statement

Generative AI was used in a limited way to refine academic phrasing, improve coherence, and ensure that complex methodological processes were communicated clearly. Its role was restricted to supporting the clarity and structure of written sections, while all research decisions, analysis, and interpretations were carried out independently. AI functioned solely as an editing and drafting aid, contributing efficiency to the writing process without influencing the intellectual content of the project.

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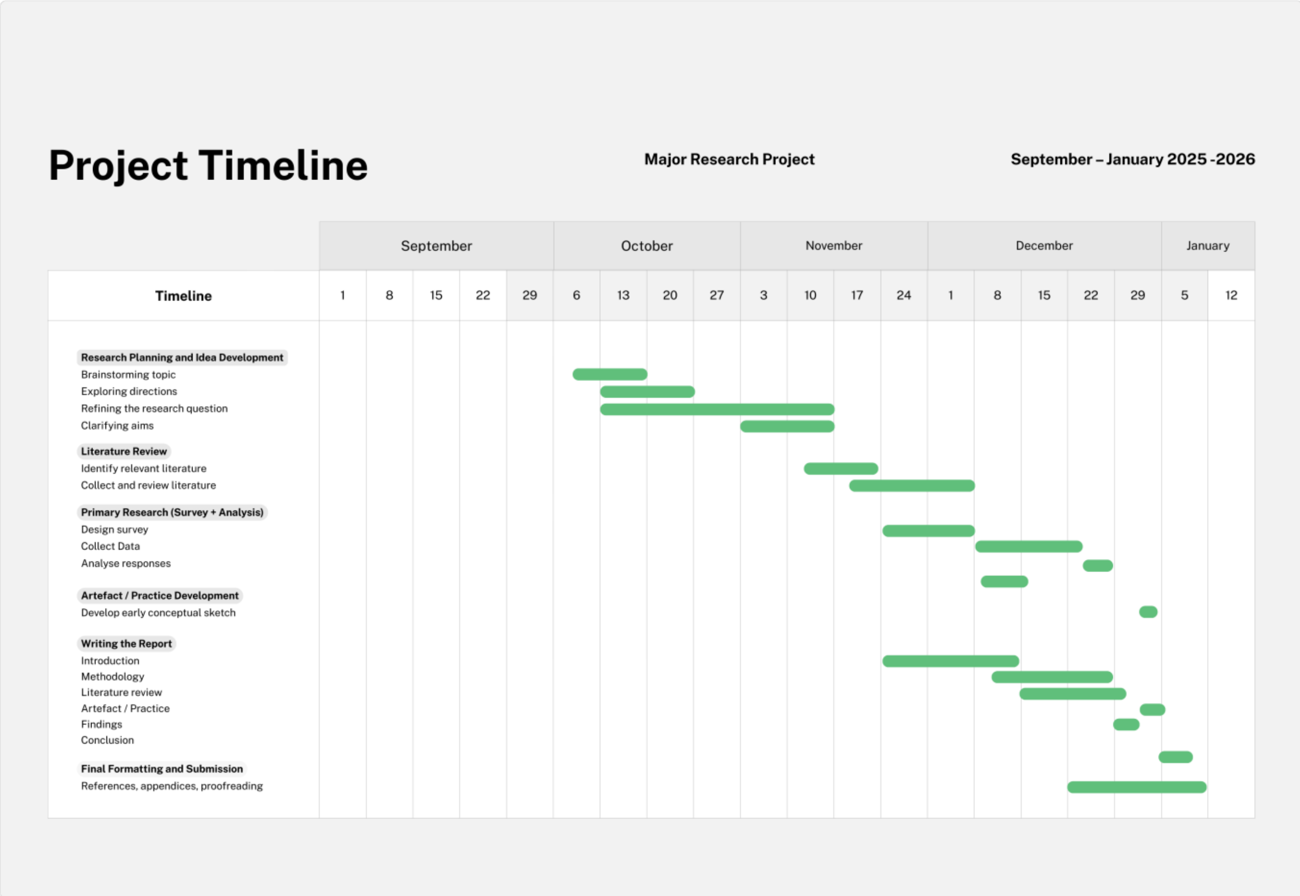
12. Appendices

Appendix A – Time Management Plan



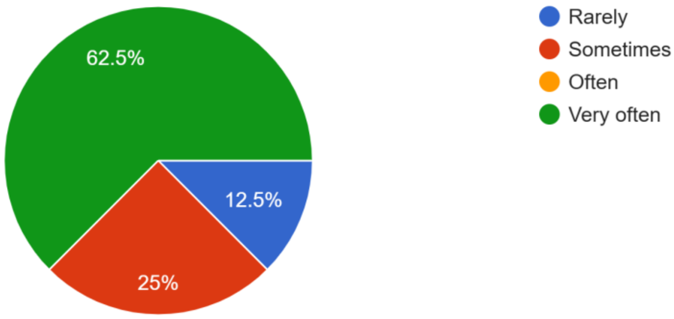
Appendix B – Questionnaire and Full Responses

How often do you engage in visually focused activities? (e.g. looking closely at images, design, photography, games, films, environments) *

- ○ Rarely
- ○ Sometimes
- ○ Often
- ○ Very often

How often do you engage in visually focused activities? (e.g. looking closely at images, design, photography, games, films, environments)

8 responses

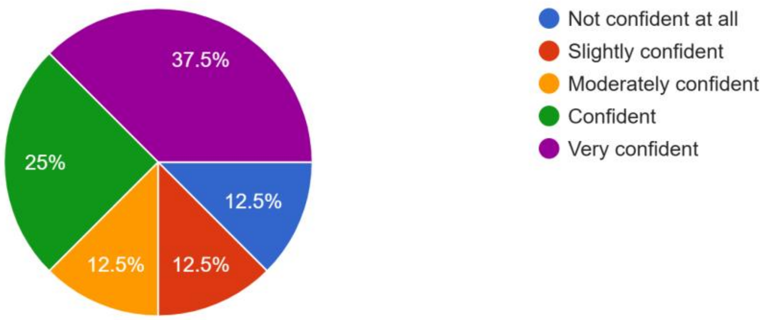


How confident are you at describing what you see? *

- ○ Not confident at all
- ○ Slightly confident
- ○ Moderately confident
- ○ Confident
- ○ Very confident

How confident are you at describing what you see?

8 responses

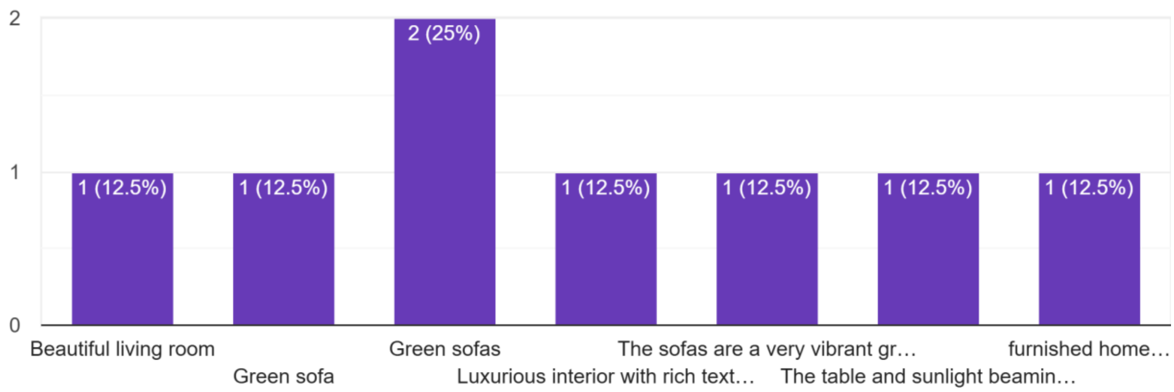


What did you notice first when viewing this scene? *

Your answer

What did you notice first when viewing this scene?

8 responses



Describe this scene in as much detail as you can. *

Your answer

Describe this scene in as much detail as you can.

8 responses

Bright green sofa, 6 dining chairs and a wooden table, coofee table; arm chair, mirrow on the wall. Everything is very modern and organised. Warm colours, good taste

Modern kitchen, very nice dining table, very bright sofas and picture

Green sofas, big open windows with long curtains, open space

It is a modern open plan kitchen/dining room/living room with a sliding door that leads to a patio. The sofas are a vibrant shade of green with a couple of white, orange and green pillows on them. The wooden table has some silver goblets and some grapes(?) with six white-grey chairs around it. The kitchen includes marble countertops and a marble island with some wood for an accent.
There seems to be a theme of nature across the room, with the colour scheme, use of wood and plants showing up in each part.

The interior is rich in textures and designed as an open-plan layout. There is a beautiful moss feature wall with subtle backlighting that works really well. The space includes thoughtful decorative elements, and from the dining area there is a clear, inviting view leading out to the terrace.

Describe this scene in as much detail as you can.

8 responses

table has some silver goblets and some grapes(?) with six white-grey chairs around it. The kitchen includes marble countertops and a marble island with some wood for an accent.
There seems to be a theme of nature across the room, with the colour scheme, use of wood and plants showing up in each part.

The interior is rich in textures and designed as an open-plan layout. There is a beautiful moss feature wall with subtle backlighting that works really well. The space includes thoughtful decorative elements, and from the dining area there is a clear, inviting view leading out to the terrace.

Sun glare; bright colours room, extra lighting in a kitchen;curtains and rounded piece of art on a wall

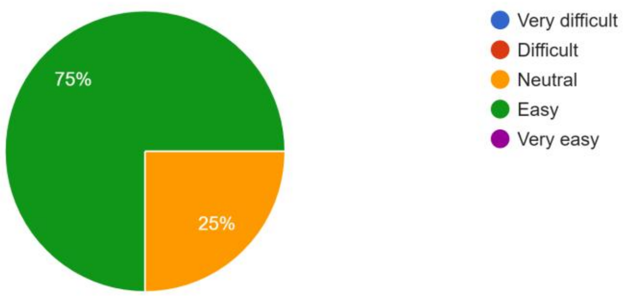
Openplanned living space, marble worktops, dining table is the focal point of the room highlighted by the light streaming in through the patio doors. Cozy seating area with emerald green sofas, small marble coffe table and high back arm chair. Natural wood floor throughout.

An open-style living space divided into three separate use areas. Living room with green sofas and a coffee table on one side of the room. Dining space in the middle of the room with a table and six chairs, with a set of direct French doors letting a lot of light in. On the opposite side of the living room, a kitchen is situated. Apart from the green sofas, most furniture is natural beige colours.

How easy was it to describe this scene? *

- ◯ Very difficult
- ◯ Difficult
- ◯ Neutral
- ◯ Easy
- ◯ Very easy

How easy was it to describe this scene?
8 responses

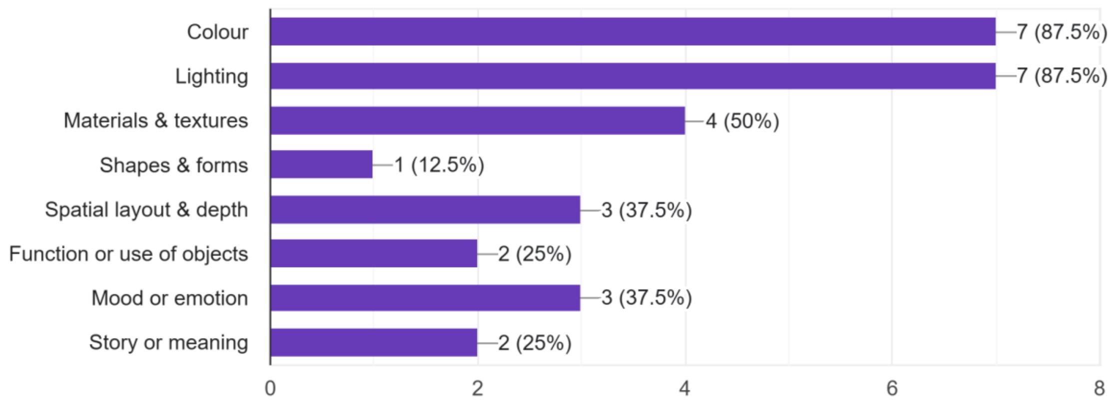


Which aspects did you focus on most while describing the scene? *

- ☐ Colour
- ☐ Lighting
- ☐ Materials & textures
- ☐ Shapes & forms
- ☐ Spatial layout & depth
- ☐ Function or use of objects
- ☐ Mood or emotion
- ☐ Story or meaning
- ☐ Other: ____________________

Which aspects did you focus on most while describing the scene?

8 responses



What did you notice first when viewing this scene? *

Your answer

What did you notice first when viewing this scene?

8 responses

Roads

Sunset

Beach, sunset, latest technology and good roads

I noticed the train on the monorail first.

Nice view

Sunrise and highway

The monorail

Light

Describe this scene in as much detail as you can. *

Your answer

Describe this scene in as much detail as you can.

8 responses

Modern future roads

Lovely evening, nice view

Futuristic scenery, modern structures, well developed city, empty roads, interesting pavements

This scene is of a street in a city at dawn, primarily using stone and asphalt as the materials. There are cycle lanes and street lights on both sides of the road. The monorail includes a train with a vibrant blue stripe (or perhaps a light) travelling on it. There is also a bunch of buildings in front of where the sun is rising, giving a glowing effect.
This scene makes me thing it takes place in the near-future.

Urban, busy area with a main road, pedestrian and cycle paths. A monorail runs through the scene, and in the distance there is a view of the sea. On the right side, some tall buildings are visible with bright light coming from that direction.

Sky, sunrise, yellow- blue-dark colours, sea; mountains; roads, straight road; shaped road, bridge; train

Describe this scene in as much detail as you can.

8 responses

cycle lanes and street lights on both sides of the road. The monorail includes a train with a vibrant blue stripe (or perhaps a light) travelling on it. There is also a bunch of buildings in front of where the sun is rising, giving a glowing effect.
This scene makes me thing it takes place in the near-future.

Urban, busy area with a main road, pedestrian and cycle paths. A monorail runs through the scene, and in the distance there is a view of the sea. On the right side, some tall buildings are visible with bright light coming from that direction.

Sky, sunrise, yellow- blue-dark colours, sea; mountains; roads, straight road; shaped road, bridge; train

Sunset scene looking over a large body of water, with what looks like monorail with train traveling along the rail and winding road in the back ground. With an empty strait road in the foreground. The streat lights are just beginning to turn on as the sun is setting. In the the very far background on the other side of the waterto the right is some tall tower buidlings. To the left is what looks like an apartment complex with a small trees

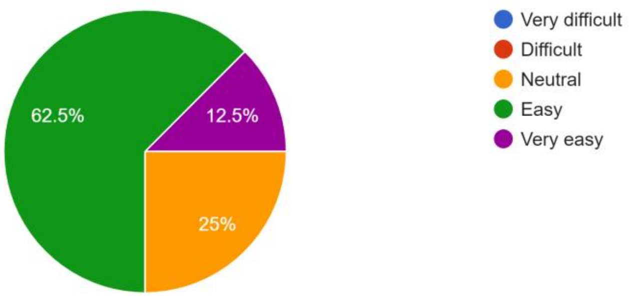
An evening scene at the waterway. The main scene in the picture is a motorway. There is a bridge and a train over the motorway. The image has a very industrial or modern feel.

How easy was it to describe this scene? *

- Very difficult
- Difficult
- Neutral
- Easy
- Very easy

How easy was it to describe this scene?

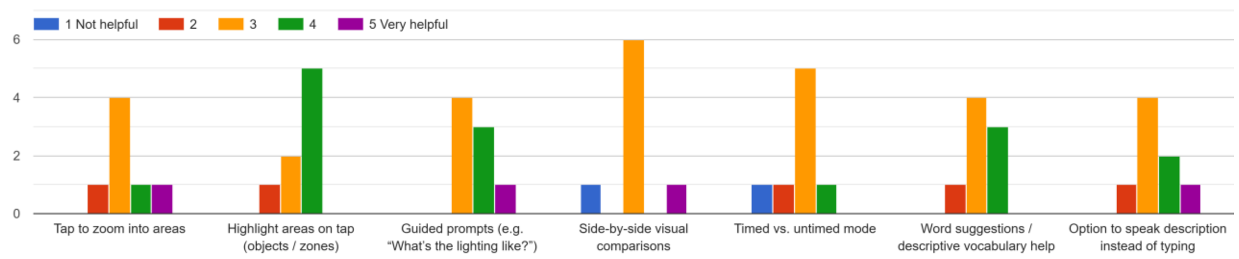
8 responses



How helpful would each of the following features be for helping you describe visual scenes?

	1 Not helpful	2	3	4	5 Very helpful
Tap to zoom into areas					
Highlight areas on tap (objects / zones)					
Guided prompts (e.g. "What's the lighting like?")					
Side-by-side visual comparisons					
Timed vs. untimed mode					
Word suggestions / descriptive vocabulary help					
Option to speak description instead of typing					

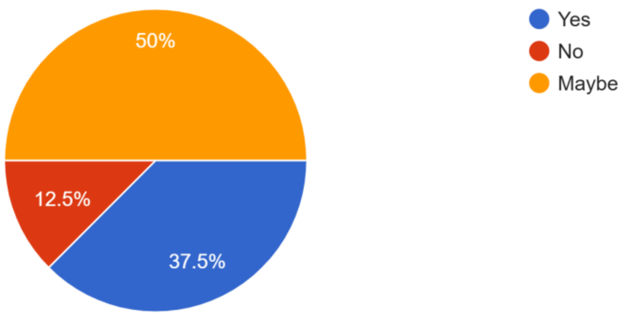
How helpful would each of the following features be for helping you describe visual scenes?



Would you use a simple mobile app to practise noticing and describing visual scenes?

- Yes
- No
- Maybe

Would you use a simple mobile app to practise noticing and describing visual scenes?
8 responses



Appendix C – Research Ethics Form 1

UNIVERSITY OF WINCHESTER

ETHICS FORM 1

WHAT LEVEL OF REVIEW DO I NEED?

GUIDELINES

This form is for staff and doctoral students. It will help you identify the level of review needed for your project. Before completing it, you need to:

1. Read *The University Research Ethics Policy*.
2. If you are a student, discuss the ethical aspects of your project with your supervisor.

It is your responsibility to follow the University's Policy on the ethical conduct of research and to follow any relevant academic guidelines or professional codes of practice pertaining to your study when answering these questions.

The questions and checklist in this proforma are intended to guide your reflection on the ethical implications of your research. Explanatory notes and further details can be found in the Policy document.

SECTION 1

DETERMINING WHETHER YOU REQUIRE ETHICS REVIEW

YOUR PROJECT
Project title: **Designing 3D Interactive Tools to Enhance Descriptive Thinking and Visual Awareness**
Your name**: Tomas Dulkys**

1.	**Is the proposed activity classified as Research or Audit /Service Evaluation or similar?**	
	☒ Research	☐ Audit or Service Evaluation
	*Use the Policy to help you answer this question. If the proposed activity meets the definition of **research** (see the policy), CONTINUE.* *If the activity is an **audit** or a **service evaluation**, STOP. You do not need to seek ethics approval, but you do need to formally register your project with UREC, along with a project outline. To do this complete Form 2.* *If you are unclear what type of activity you are undertaking, please refer to the Policy for additional types.*	
2.	**Does the research involve living human participants, human samples or data derived from individuals who may be identifiable through the data collected?**	
	☒ Yes	☐ No
	Use the Policy to help you answer this question. *If you answer **NO**, SKIP to QUESTION 6 and CONTINUE.* *If you answer **YES**, CONTINUE.*	
3.	**Is the research being conducted for a medicinal purpose?**	
	☐ Yes	☒ No
	Use the Policy to help you answer this question. See Appendix 2 - FAQs and definitions. *If you answer **YES**, and think your research comes under the definition of 'for a medicinal purpose,' it will need to be scrutinised by the Committee. Please email the Committee Chair (ethics@winchester.ac.uk) for further guidance on what to do.* *If you answer **NO**, CONTINUE.*	
4.	**Does your research require external ethics approval or review?**	

	☐ Yes	☒ No
	For example, from the NHS or another overseeing body. Use the Policy to help you answer this question. *If you answer **NO**, CONTINUE.* *If you answer **YES**, you need to formally register your project with UREC, along with the relevant external ethics approval. To do this complete Form 2.*	
5.	**Is the project underway and, the researcher or PI, has moved institution to Winchester?**	
	☐ Yes	☒ No
	*If you answer **YES**, please read the following:* *If the research began when the PI was employed at another institution but has subsequently moved to Winchester, and the project has previously been subjected to ethics scrutiny at that institution, then it need not go through ethics review again. The outcome of ethics review by that institution should be communicated to UREC for formal recording. To do this complete Form 2 and include evidence of the previous ethics approval.* *HOWEVER, if there have been significant changes to the original research design which have ethical implications or recruitment of a cohort of participants will be undertaken through Winchester, then the project will require ethics review and you should apply for approval, CONTINUE.* *If you answer **NO**, CONTINUE.*	
6.	**Is the research collaborative?**	
	☐ Yes	☒ No
	*If you answer **YES**:* • *where the Principal Investigator (PI) of the research is located at another institution, it is their responsibility to seek ethics approval, including partner research sites. The outcome of ethics review by that institution should be communicated to UREC for formal recording. To do this complete Form 2 and include evidence of the previous ethics approval.* • *where the PI is located at Winchester, then the project will undergo scrutiny as per Winchester's Ethics Policy, CONTINUE.* *If you answer **NO**, CONTINUE.*	
7.	**Is the research being conducted in another country?**	
	☐ Yes	☒ No
	*If you answer **YES**, please read the following:* *Where a project is conducted in another country, the researcher should consider if it is possible to obtain ethics review by a local research ethics committee or other relevant*	

	body. The outcome of such a review by that institution should be communicated to UREC for formal recording, along with a project outline. To do this complete Form 2. *If this is not possible, the project should be reviewed by the University of Winchester, either at Faculty level or Committee depending on the nature of the proposed work, so CONTINUE.*	
8.	**Does the research involve the use of documentary material(s) for analysis - for example artifacts, papers, historical records, literary works or documents in a public or private archive?**	
	☐ Yes	☒ No
	Note: Documentary material does not include academic papers or other 'building block' literature in the public /academic domain which is used to inform the research context or rationale for the study. Instead, the documentary material would be the 'data' for the study, therefore literature reviews or literature critiques are not considered documentary research. *If you answer **YES**, you need to formally register your project with UREC, along with a project description. To do this complete Form 2. Where materials are in a private archive or closed collection, please include details of the nature of the private archive /closed collection and provide evidence of permission to use this material for research purposes. Please also consider if there may be outcome ethical implications e.g. the subject matter may have a negative impact on those still connected to the materials.*	
9.	**Does the research involve live vertebrate animals?**	
	☐ Yes	☒ No
	*If you answer **NO**, CONTINUE.* *If you answer **YES**, you need to formally register your project with UREC, along with a copy of the relevant licence (if required). To do this complete Form 5.*	
10.	**Does the research involve environmental interventions?**	
	☐ Yes	☒ No
	*If you answer **NO**, CONTINUE.* *If you answer **YES**, you need to formally register your project with UREC, along with a copy of the relevant licence (if appropriate). To do this complete Form 2*	
11.	**Does the project pose any potential or actual conflict(s) of interest for the researcher and /or stakeholders?**	
	☐ Yes	☒ No
	*If you answer **YES**, please ensure you provide information on the form you complete.*	
12.	**Does the data you will collect contain *any* information that could be linked back to participants or that might identify them (e.g. name, address, photo, voice, email)?**	

	☐ Yes	☒ No
	*If you answer **NO**, you need to formally register your project with UREC. To do this complete Form 2.* *If you answer **YES**, CONTINUE.*	

☞ Reaching the end of these questions, **either** you will have been directed to complete a specific additional form **or** you should continue to section 2.

If you are still unsure whether you need ethics review or not, please re-read The Policy and email your query to ethics@winchester.ac.uk with details of your project.

	Please mark with an ☒ as appropriate	YES	NO
1.	Does the research involve individuals who might be considered vulnerable? *For example: vulnerable children, over-researched groups, people with learning difficulties, people with mental health problems, young offenders, people in care facilities, including prisons. For a note on research with children, see Appendix 2 of the Policy.*	☐	☒
2.	Does the research involve individuals in unequal relationships e.g. your own students? *Please note:* 1. *students recruited via SONA are not considered 'your own students.' If you intend to recruit widely across the University or your Faculty (e.g. through snowball sampling or a mail shot) you do not need to consider such students as your own, even if some participants may be students you are directly involved with. Only tick "yes" if you are targeting your own students specifically.* 2. *if you are an undergraduate or postgraduate student carrying out research with children in either a school or early years setting, these DO NOT come under the category of your 'own students.'*	☐	☒
3.	Will it be necessary for participants to take part in the study without their knowledge and consent at the time? *For example: covert observation of people in non-public places, use of deception. See Appendix 2 of the Policy.*	☐	☒
4.	Will the study involve discussion of sensitive or personal topics? *For example: (but not limited to) participants' relationships, emotions, sexual behaviour, experience of violence, mental health, gender, race / ethnicity status or experience, political or religious affiliations. Please refer to the Policy.*	☐	☒
5.	Is there a risk that the highly sensitive nature of the research topic might lead to disclosures from the participant concerning their own involvement in illegal activities or other activities that represent a threat to themselves or others which may need onward reporting? *For example: sexual activity, drug use, illegal activities or professional misconduct.*	☐	☒
6.	Might the research involve the sharing data or confidential information beyond the initial consent given?	☐	☒

7.	Might participant anonymity be compromised at any time during or after the study? *For example: will the research involve respondents using the internet, social media, or other visual /vocal methods where respondents may be identified?*	☐	☒
8.	Is the study likely to induce severe physical harm or psychological distress?	☐	☒
9.	Does your research involve tissue samples covered by the Human Tissue Act (2004)?	☐	☒
10.	Is there a possibility that the safety of the researcher may be in question? *For example: research in high-risk locations or with high-risk groups.*	☐	☒
11.	Does the research involve creating, downloading, storing or transmitting material that may be considered to be unlawful, indecent, offensive, defamatory, threatening, discriminatory or extremist? *If you answer **YES** to this question, you must also contact the Director of Library and IT Services, who must provide approval for the use of such data.*	☐	☒

SECTION 2

DETERMINING THE LEVEL OF ETHICS REVIEW REQUIRED

Answering **NO** to ***all*** these questions means your project is eligible for Faculty level ethics review.
You now need to complete Form 3.

Answering **YES** to ***any*** of these questions means your project will require Committee ethics review.
You now need to complete Form 4.

Appendix D – Research Ethics Form 3

ETHICS FORM 3

FACULTY REVIEW

GUIDELINES

This form is for staff and doctoral students. It will help you set out the ethical aspects of your project that need to be reviewed. Before completing it, you need to:

3. Read *The University Research Ethics Policy*.
4. If you are a student, discuss the ethical aspects of your project with your supervisor.

It is your responsibility to follow the University's Policy on the ethical conduct of research and to follow any relevant academic guidelines or professional codes of practice pertaining to your study when answering these questions. This includes providing appropriate information sheets and consent forms and ensuring confidentiality in the storage and use of data.

The questions in this proforma are intended to guide your reflection on the ethical implications of your research. Explanatory notes and further details can be found in the Policy document.

If any aspect of your project changes during the course of the research, you must notify the Chair of UREC.

SECTION 1

YOUR DETAILS			
1.1.	Your name**: Tomas Dulkys**		
1.2.	Your department: School of Creative Performance and Production		
1.3.	Your Faculty: Education and Arts		
1.4.	Your status**:**		
		☒ Undergraduate Student	☐ Staff (Professional Services)
		☐ Taught Master	☐ Staff (Academic)
		☐ Research Degree Student	☐ Other (please specify below)
1.5.	Your university email address**: T.Dulkys.21@unimail.winchester.ac.uk**		
1.6.	Your telephone number**: 07599639558**		
	<u>**For doctoral students only**</u>:		
1.7.	Your degree programme**:**		
1.8.	Your supervisor's name**:**		
1.9.	Your supervisor's department**:**		
1.10.	Your supervisor's email**:**		

SECTION 2

YOUR PROJECT			
2.1.	Project title**:** Designing 3D Interactive Tools to Enhance Descriptive Thinking and Visual Awareness		
2.2.	Start date**:** 14/011/2025		
2.3.	Expected completion date**:** 04/01/2026		
2.4.	Expected location of data collection**:** (*e.g. school, workplace, public place, University premises etc*.)		
2.5.	Has funding been sought for this research**?**		
		☐ Yes	☒ No
2.6.	If so, where have you applied for funding?		
2.7.	Has the funding been granted?		
		☐ Yes ☒ No	☐ Pending
2.8.	Is the research collaborative? (*e.g. co-investigators from another institution, at or with another organisation or colleagues in another department*)		
		☐ Yes	☒ No
		If yes, which?	
2.9.	Is Disclosure and Barring Service clearance required for your study? *It is your responsibility to contact the Disclosure and Barring Service (DBS) to confirm whether or not clearance is needed prior to commencing recruitment or data collection. More information here*		

		☐ Yes	☒ No
2.10.	Is a risk assessment required? *It is your responsibility to contact the Health and Safety Office at the University to confirm whether or not a risk assessment is required prior to commencing recruitment or data collection.*		
		☐ Yes ☒ No	☐ Pending
2.11.	Will your research be informed by guidelines from a professional association or specific, agreed standards of practice?		
		☐ Yes	☒ No
		If yes, which?	

SECTION 3

PROJECT DESCRIPTION
Research Title: *Designing 3D Interactive Tools to Enhance Descriptive Thinking and Visual Awareness* **Research Question:** *How can 3D-based interactive visual design enhance descriptive thinking and visual awareness in adults?* **Rationale** In daily life, many people find it difficult to describe what they see. When viewing a landscape, a painting, or even a film scene, they might simply say, "That's beautiful," without being able to explain why. This limited descriptive ability can reduce creativity, confidence, and visual awareness in both everyday life and professional contexts. The idea for this project came from personal reflection and from my professional interest in 3D visualisation and digital design. I realised that developing tools to help people notice and describe visual details could also improve imagination and creative communication. Research in psychology and design supports the importance of visual perception and description. Rudolf Arnheim (1969) and Barbara Tversky (2011) explain that perception and imagination are closely linked, and that describing what we see strengthens our understanding of form, space, and emotion. In design education, Goldschmidt (2017) and Lawson (2004) show that verbalising visual ideas helps designers think more creatively and reflectively. From a user-experience perspective, Norman (2013) and Shneiderman (2022) argue that well-designed interfaces can improve attention, engagement, and learning. Studies on spatial and 3D visualisation (Billinghurst & Duenser, 2012) suggest that interactive environments can improve focus and visual awareness. Despite this, most existing digital tools—such as Lumosity, CogniFit, and BrainHQ—mainly train speed, memory, or logic rather than descriptive or observational thinking. This project aims to address that gap by exploring how 3D-based interactive design can be used to strengthen adults' ability to observe, describe, and reflect on visual information. **Aims** 1. To analyse key theories and studies on descriptive thinking, visual awareness, and 3D interaction design. 2. To investigate how adults perceive and describe visual details through a short survey and examples of visual scenes.

3. To translate findings into design ideas that guide prototype development for an interactive visual learning app in Term 2.

Methods

This research will follow a qualitative mixed-method approach. A literature review will examine academic studies on imagination, perception, and interactive visual design to define the theoretical background. A short online survey will collect feedback from adult participants (ages 18+), exploring how they describe and interpret visual examples such as images or 3D renders. These data will be thematically analysed to identify patterns in descriptive language, focus, and engagement. Findings will inform early design sketches and conceptual wireframes for a potential mobile-app-based interactive experience, to be further developed in the second term.

Ethical Considerations

Participants will be fully informed about the study's purpose before completing the survey. Participation will be voluntary and anonymous, with no sensitive personal data collected.

SECTION 4

REFINING THE LEVEL OF ETHICS REVIEW REQUIRED

Please mark with an ☒ as appropriate		**YES**	**NO**
1	Does the research involve members of the public in a research capacity as co-researchers? (I.e. as in participant research where involvement extends beyond data collection)	☐	☒
2	Is there a risk of over-disclosure that may put the participants at risk or cause them any anxiety?	☐	☒
3	Will tissue samples (including blood) be obtained from participants?	☐	☒
4	Will the study require the co-operation of a gatekeeper for initial access to participants? (E.g. to students at school, to members of self-help group.)	☐	☒
5	Is the right to withdraw from the study withheld at any time, or not made explicit?	☐	☒
6	Is there any reason participants may feel obliged to participate in the study against their will?	☐	☒
8	Will the research involve administrative or secure data that requires permission from the appropriate authorities before use?	☐	☒
10	Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?	☐	☒
11	Are there payments to researchers /participants that may have an impact on the objectivity of the research?	☐	☒
12	Is there any cause for uncertainty as to whether the research will fully comply with the requirements of the General Data Protection Regulation (GDPR) (2018)?	☐	☒
13	Does any part of the project breach any codes of practice for ethics in place within the organisation in which the research is taking place?	☐	☒

14	Are drugs, placebos or other substances (e.g. food substances, vitamins) to be administered to the study participants? Please note: for fast track review, it is expected that the study will not involve invasive, intrusive or potentially harmful procedures of any kind.	☐	☒
15	Is pain or more than mild discomfort likely to result from the study?	☐	☒
16	Could the study induce psychological stress or anxiety or cause harm or negative consequences beyond the risks encountered in normal life? (E.g. involve prolonged or repetitive testing.)	☐	☒
17	Does the project pose any potential or actual conflict(s) of interest for the researcher and /or stakeholders?	☐	☒

If you answer YES to *any* of these questions, please use the next section to indicate which question you have said yes to, describe the ethical issue in the context of your study and how you will address it. If you have answered NO to all questions, complete section 6.

SECTION 5

ADDITIONAL INFORMATION AND AMENDMENTS
Use this space to address ethical issues highlighted by the checklist in section 4, or to amend an original submission.

SECTION 6

DECLARATION
I have read and understood the University of Winchester Research Ethics Policy and confirm that adequate safeguards in relation to the ethical issues raised by this research can and will be put in place. I am aware of and understand University procedures regarding Health and Safety. I understand that the ethical aspects of this project may be monitored by the University Research Ethics Committee. I understand my responsibilities as a researcher as described in the University of Winchester Research Ethics Policy. I declare that the answers above accurately describe the research as presently designed and that a new application will be submitted should the research design change in a way which would alter any responses given in Form 1 or here.
☒ I confirm that if a Risk Assessment is required I will complete it and have it co-signed by my Supervisor or Head of Department before data collection takes place.
☒ I confirm that, if DBS clearance is required for my project, then I will seek it before the start of my project.
☒ I confirm that my research does not include risks that might cause it to be excluded from coverage by the University's insurers.

☒ I confirm that I have appropriate insurance for this research.	
Researcher’s signature: Tomas Dulkys	Date: 12/11/2025
In addition, for **students** (research): The student has the skills to carry out the proposed research. I undertake to monitor the student’s adherence to the relevant research guidelines and codes of practice.	
Supervisor’s signature:	Date:

Please submit this form along with Form 1 to your nominated Ethics Lead.

Please remember to append any forms or documents that may be relevant to your application (e.g. consent form, information sheet, questionnaire(s) etc.). Your form cannot be considered unless it is submitted with the required supporting documentation. Omitting to do so will delay the ethics review process.